Purrfect Stay

Pet Hotel Management System

Final Report

CN230 - Database Systems
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1. Introduction

1.1 System Overview

The pet boarding industry is built on trust, yet conventional facilities predominantly depend on fragmented, paper-based workflows, physical folders, or rudimentary spreadsheets to administer their daily operations. In a bustling pet hotel environment, this manual approach inevitably introduces severe operational bottlenecks and critical errors. For instance, front-desk staff utilizing spreadsheets often face concurrent room reservation conflicts, leading to embarrassing double-bookings during peak holiday seasons. On the facility floor, caretakers struggle with misplaced or outdated physical vaccination documentation, and critical miscommunications frequently occur regarding a specific pet's dietary allergies or medication schedules. Furthermore, pet owners are generally left in the dark, afforded limited to no real-time visibility into their animals' well-being during the boarding period. This lack of transparency and operational friction not only causes daily stress for the staff but ultimately diminishes customer satisfaction and erodes the foundational trust required in pet care.

The Purrfect Stay system was conceived to systematically eliminate these industry shortcomings by serving as the central digital nervous system for a modern pet boarding facility. It is a comprehensive, database-driven Pet Hotel Management System that digitally transforms end-to-end facility operations. Rather than treating bookings, pet care, and billing as separate tasks, Purrfect Stay unites them into a single, cohesive ecosystem. When a receptionist books a room, the system instantly updates real-time availability to prevent overlaps, locks in the financial rates to ensure billing accuracy, and immediately generates a digital profile for the caretakers. The system strictly enforces data integrity—for example, it makes it impossible to check a pet in without explicitly verifying their allergy and medical condition statuses. It empowers caretakers to log structured daily reports (tracking food intake, potty status, and mood) which form a transparent care history. Ultimately, Purrfect Stay replaces administrative guesswork with automated workflows, allowing the staff to focus less on paperwork and more on providing high-quality care to the animals.

1.2 Project Goals and Scope

The primary objective is to develop a robust relational database system capable of supporting the complex operational logic of a modern pet boarding facility. The scope covers:

- Customer and Pet Profiling: Managing owner contacts and comprehensive pet medical histories.
- Reservation Management: Real-time room availability verification and service scheduling.
- Billing and Invoicing: Automated computation of fees and payment tracking.
- Daily Care Monitoring: Logging feeding, behavior, and observations.

Inventory Control: Managing essential facility supplies and pet food stock.

2. System Analysis

2.1 Stakeholders and Their Roles

The system identifies several primary stakeholders who interact directly with the application:

- Pet Owner (Customer): Registers accounts, manages pet profiles, makes reservations, selects add-on services, and reviews daily care reports.

- Receptionist (Front Desk): Manages the booking pipeline, verifies credentials during check-in, processes check-outs, and calculates final invoices.

- Pet Caretaker: Logs daily operational data (feeding, mood, behavior), executes meal plans, and confirms service completion.
- General Staff: Handles facility maintenance and requires access to operational schedules and room statuses.

- Hotel Manager: Monitors room utilization, tracks inventory levels, and manages staff accounts
  .
- Business Owner: Requires access to high-level strategic data, financial summaries, and audit trails for transparency.

2.2 Use Cases

The system includes thirty-one use cases categorized into six functional modules:

- Module 1: Booking & Front Desk Management
  - 1.1 Room Reservation: Selecting dates, viewing availability, and confirming bookings.
  - 1.2 Check-In and Check-Out Processing: Verifying information and updating room occupancy.
  - 1.3 Add-On Service Management: Appending services like grooming to active bookings.

- Module 2: Billing & Payment Management
  - 2.1 Bill Calculation and Invoice Generation: Aggregating room fees and service charges.
  - 2.2 Payment Recording and Receipt Issuance: Confirming payments via various methods and issuing receipts.

- Module 3: Pet Care & Daily Reporting
  - 3.1 Daily Care Report Logging: Recording feeding, mood, and behavioral data with photos.
  - 3.2 Meal Plan Configuration: Defining food types and portion quantities for each pet.

- Module 4: Inventory & Business Management
  - 4.1 Food Stock Management: Retrieving inventory lists and managing restocks.
  - 4.2 Supplies Inventory Management: Filtering items by category and tracking stock thresholds.
  - 4.3 Business Overview Dashboard: Visualizing revenue and room utilization statistics.

- Module 5: Staff & Access Control Management
  - 5.1 Staff Profile and Role Management: Creating and assigning roles to staff accounts.
  - 5.2 System Authentication: Role-based login and logout procedures.
  - 5.3 Attendance Time Logging: Recording daily clock-in and clock-out timestamps.
  - 5.4 System Audit Trail Review: Reviewing logs of critical data modifications.

- Module 6: User Account & Profile Management
    - 6.1 Customer Account Registration: New account creation with OTP verification.
    - 6.2 Pet Profile Registration: Recording pet details, health notes, and vaccination history.
    - 6.3 Notification Management: Dispatching automated alerts for care reports and booking updates.

2.3 Functional Requirements

The system fulfills twenty-six functional requirements (FR1–FR7):

- FR1 (User & Access): OTP verification for customers, role-based dashboards, manager-level staff control, attendance tracking, and audit trails.

- FR2 (Pet Profile): Creation of pet profiles, vaccination records with PDF support, medical notes, and individualized meal plans.

- FR3 (Booking): Real-time availability, automatic fee calculation with price locking, check-in/out functions, and booking cancellations with identity tracking.

- FR4 (Pet Care): Daily health reporting for caretakers and automated notifications for pet owners.

- FR5 (Billing): Consolidated invoice generation and multi-method payment recording.

- FR6 (Inventory): Automatic stock deduction and low-stock alerts at 20% thresholds.

- FR7 (Notifications): Immediate owner notifications for status changes and receptionist alerts for new bookings

3 Proposed Technology Stack

The Purrfect Stay system employs a robust, multi-tiered architecture designed to prioritize data integrity, security, and performance. The technology selections emphasize explicit control over data operations and a decoupled structure for future scalability.

- Database Tier (PostgreSQL & Supabase): The system utilizes PostgreSQL as its primary Relational Database Management System (RDBMS) for its full ACID compliance, which is essential for managing critical financial and booking transactions. The database is hosted on Supabase, providing a cloud-native environment for high availability. A key design choice was the avoidance of heavy Object-Relational Mappers (ORMs) in favor of the psycopg2 adapter, allowing for the explicit execution of raw SQL queries to maximize performance and ensure academic transparency. The schema further leverages PostgreSQL's advanced features, such as custom ENUM types for status consistency and generated columns for automated, immutable billing calculations.

- Backend Tier (Python & Flask): The application logic layer is built with Python 3.10 and the Flask micro-framework. To ensure a clean and maintainable codebase, the backend is organized into a Blueprint architecture, separating logic into modular services such as authentication, pet profiling, and analytics. Security is a primary focus, utilizing bcrypt for salted password hashing and PyJWT for stateless, token-based authentication (JSON Web Tokens), ensuring secure access control for staff and administrators.

- Presentation Tier (Decoupled Frontend): The user interface is constructed using HTML and Tailwind CSS, providing a utility-first approach to responsive design. The frontend follows a modern, decoupled architecture where client-side logic—organized into shared components like sidebar and navbar—interacts with the backend via a RESTful API provider. This structure allows the interface to update dynamically without full page reloads, enhancing the operational speed for hotel staff.

- Development & Infrastructure Tools:
    - Docker: Used for containerization to ensure consistency between development and deployment environments.
    - Git & GitHub: Employed for version control and collaborative management via GitHub Projects' Kanban tracking.
    - Figma & Canva: Utilized for high-fidelity UI wireframing and the creation of architectural diagrams and visual assets.

4. Database Design (Comprehensive Overview)

The architectural design of the Purrfect Stay database is built upon a multi-layered relational framework, specifically engineered to mirror the intricate operations of a high-standard pet boarding facility.
The design strategy focuses on three core structural pillars:

- Hierarchical Entity Relationships: At the foundational level, the database establishes a strict parent-child relationship between customers and their pets. This ensures that every medical record, dietary preference, and behavioral note is permanently anchored to a verified owner, facilitating long-term care history and personalized service delivery.

- Transactional Stay Management: The system's core logic revolves around the "Stay" as the primary unit of data. By mapping specific pets to individual rooms through a dedicated association layer, the database effectively manages spatial resources. This approach allows the hotel to prevent room conflicts while simultaneously tracking all services—from standard grooming to specialized medical attention—directly to the active boarding event

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- Modular Care and Financial Integration: The database is segmented into functional modules that handle pet monitoring, resource logistics, and financial closure independently yet cohesively. This modularity ensures that care logs for health tracking, inventory usage for supply management, and billing for revenue recording all interact within the same ecosystem, providing a 360-degree view of every pet's stay without data duplication.

- Systemic Integrity through Categorization: The design utilizes a robust categorization system for room types, pet species, and operational statuses. By pre-defining these categories, the database enforces a high degree of data consistency, ensuring that analytical reports accurately reflect the diversity of the hotel's residents and their specific needs.

5. Database Development (Strategic Operational Logic)

The development phase focused on transforming high-level business rules into automated database-level safeguards, ensuring operational reliability and financial transparency.

- Lifecycle and State Transition Logic: The development enforces a standardized lifecycle for every reservation, guiding it through a sequence of verified states—from initial request to active boarding and final completion. This transition logic prevents operational errors, such as checking a pet out before they have officially checked in or allowing multiple pets to occupy a room beyond its capacity.

- Financial Safeguarding and Rate Integrity: A critical feature of the development is the implementation of price-locking mechanisms. To ensure trust and transparency, the database captures and secures the agreed-upon rates at the exact moment of booking confirmation. This safeguards the transaction against future price fluctuations, ensuring that the financial commitment remains consistent throughout the duration of the stay.

- Operational Safety and Care Compliance: To uphold high standards of pet care, the system is developed with mandatory reporting requirements. For every pet currently staying at the facility, the database requires specific health and behavioral inputs to be recorded daily. This ensures that caretakers are held accountable and that owners receive a complete and accurate history of their pet's well-being.

- Resource Management and Alert Systems: The development includes intelligent monitoring of facility resources. By tracking the consumption of food and supplies during each stay, the system provides real-time visibility into stock levels. This proactive approach allows the management to identify low-stock situations before they impact daily operations, ensuring continuity of care.

- Immutable Tracking and Security: To ensure professional accountability and data security, a comprehensive audit layer was developed. This layer silently records all critical modifications within the system, providing an unalterable history of administrative actions. This is essential for resolving discrepancies and maintaining a transparent operational environment for both management and owners.

6. Key Queries & Operational Database Logic

The system relies on advanced relational database operations to process complex business rules directly at the data layer. This design significantly reduces the computational burden on the backend server and ensures high-speed data delivery to the frontend. The following outlines the core logical operations powering the system's most critical features

6.1 Executive Revenue & Booking Summary

This operation acts as the primary data engine for the management dashboard, designed to compile a comprehensive overview of the hotel's daily performance in a single database execution.

- Operational Detail: Rather than forcing the backend application to make multiple separate requests (e.g., asking the database once for active stays, again for completed stays, and again for revenue), the database utilizes a technique known as conditional aggregation. It scans the entire booking ledger within a user-defined date range and categorizes every reservation simultaneously.

- Financial Accuracy: To calculate total revenue, the operation intelligently links booking records to their corresponding financial invoices. It applies a strict filtering rule to only sum invoices that have been officially marked as "Paid" or "Partially Paid." Furthermore, it incorporates a mathematical fail-safe that guarantees a clean "zero" is returned if no transactions occurred during the period, preventing potential calculation errors or system crashes on the frontend.

6.2 Daily Care Fulfillment Tracking

This query operates as a real-time compliance and safety monitor for the pet care module. Its primary objective is to ensure that every boarded pet receives their mandatory daily health assessment without fail.

- Operational Detail: To maximize system performance, this operation does not load the heavy, historical data of a pet's past care logs. Instead, it utilizes a highly optimized boolean verification method (a simple "true/false" check). The database first isolates the list of pets currently occupying a room today. It then rapidly scans the care log table to verify if at least one health report has been submitted for each specific pet on the current date.

- Timezone Synchronization: Because cloud servers often run on Coordinated Universal Time (UTC), this database operation automatically casts all timestamps to the local time zone. This guarantees that the "daily deadline" for staff reporting aligns perfectly with the facility's actual operating hours.

6.3 Comprehensive Booking Profiling (Front Desk Hub)

This operation serves as the central data hub for the reception staff, compiling highly fragmented relational data into one cohesive, easy-to-read reservation profile.

- Operational Detail: A single pet's stay involves data scattered across multiple tables: customer contact details, pet health profiles, physical room assignments, stay durations, and additional service requests. This operation seamlessly weaves these five distinct data sources together before delivering them to the user interface.

- Intelligent Data Grouping: A major technical achievement in this operation is its handling of add-on services. If a customer requests multiple services for a single stay (such as a spa bath and nail clipping), standard database joins would generate messy, duplicate rows for the same booking. To solve this, the database intelligently collapses these multiple services into a single, structured list attached to the main reservation. This completely eliminates data redundancy and allows the frontend to render a clean, professional booking table.

7. Assumptions & Challenges

Design Assumptions

The following assumptions have been established to define the operational boundaries of the system and to guide all subsequent design and development decisions.

- Single Pet per Booking Record Each booking record is associated with one pet only. In cases where a single owner wishes to board multiple pets simultaneously, the system shall generate separate and independent booking records for each individual pet.

- Daily-Level Room Availability Management Room availability is managed at the granularity of a full calendar day. The system does not support hourly or time-slot-based booking. Check-in and check-out are processed on a per-day basis.

- Internal Staff Access Only In the initial release phase, system access is restricted exclusively to internal staff, including receptionists, pet caretakers, general staff, managers, and the business owner. Customer-facing login and self-service functionalities are not included within the current project scope and are designated for a future development phase.

- Free-Text Medical and Vaccination Records Vaccination histories and medical information are stored as free-text notes entered manually by staff. The system does not implement a structured medical record schema in this phase. Attachment of veterinary certification documents in PDF format is supported as supplementary evidence.

- External Payment Processing The system does not integrate with any payment gateway or financial processing service. All monetary transactions are handled through external means (e.g., cash, QR PromptPay, credit card terminal). The system records only the payment status (Paid / Unpaid) and the payment method for billing and reporting purposes.

- Fixed Room Rate at Time of Booking The total room charge is calculated and locked at the time of booking confirmation based on the room's base rate at that moment. Subsequent modifications to the room's base rate shall not retroactively affect any previously confirmed booking records.

Expected Challenges and Limitations

- Concurrent Booking Conflicts and Race Conditions When two or more staff members attempt to reserve the same room for an overlapping date range

simultaneously, the system is susceptible to double-booking errors. This challenge requires the implementation of transaction-level locking mechanisms within the PostgreSQL database — specifically, the use of SELECT FOR UPDATE or serializable transaction isolation — to ensure that room availability checks and reservation insertions are executed as atomic operations.

- Data Completeness and Input Quality Pet medical histories, vaccination records, feeding preferences, and behavioral notes are entered manually by staff. Inconsistent or incomplete data entry represents a persistent risk to data quality. While the system enforces required fields and basic validation rules, these measures cannot fully guarantee the accuracy or completeness of free-text inputs. Staff training and standardized data entry protocols will be essential to mitigating this limitation.

- Scope Creep and Feature Boundary Management Several features were discussed informally during the requirements gathering phase, including customer-facing online booking, automated SMS and email reminder notifications, LINE integration, and a customer loyalty rewards program. These features carry significant implementation complexity and are explicitly excluded from the current release scope. Without clearly defined boundaries, incremental feature additions risk destabilizing the project timeline and compromising the quality of core deliverables.

- Notification Delivery Reliability The system relies on external notification channels — particularly LINE — for dispatching daily care report updates, booking confirmations, and reminder alerts to pet owners. Delivery reliability is subject to third-party service availability and network conditions, which fall outside the system's direct control. Failures in notification delivery may affect customer satisfaction and trust in system responsiveness.

8. Project Plan (4-Week Implementation)

The project was successfully executed through six compressed phases over 4 weeks:

Phase	Activities	Duration	Status
Phase 1: Requirements	Finalized use cases and functional requirements.	Week 1	Completed
Phase 2: DB Design	Normalized schema to 3NF and defined all constraints.	Week 2	Completed
Phase 3: Development	Implemented PostgreSQL schema and Flask backend with raw SQL.	Week 3	Completed
Phase 4: QA & Testing	Conducted unit tests on SQL queries and finalized documentation.	Week 4	Completed

9. Backend Services Overview

The Purrfect Stay backend acts as the central engine of the application, securely managing all data processing and business logic. Instead of direct database manipulation, the frontend communicates with a structured service layer that handles the following operational capabilities

9.1 Authentication & User Management Services

This module manages system security and user identities, ensuring that only authorized personnel can access sensitive hotel data.

- Access Control: Securely verifies staff credentials and issues encrypted session tokens.
- Role Management: Ensures that staff members only access data and functions permitted by their specific roles (e.g., Caretaker vs. Manager).
- Attendance Tracking: Logs daily staff clock-in and clock-out events.
- Account Creation: Facilitates the secure registration of new pet owner accounts.

9.2 Booking & Front Desk Services

This module serves as the operational core for receptionists, handling the complete lifecycle of a reservation.

- Availability Engine: Instantly cross-references requested dates against current room occupancy to provide real-time availability.
- Reservation Processing: Secures room assignments, records associated pets, and locks in the base rate at the time of booking.
- Stay Lifecycle Management: Updates the status of a reservation from pending, to active (check-in), and finally to completed (check-out) or cancelled.
- Service Upgrades: Allows staff to seamlessly add extra chargeable services (like grooming) to an active stay.

9.3 Pet Care & Health Monitoring Services

Dedicated to maintaining high standards of animal welfare and transparent record-keeping.

- Profile Management: Stores and retrieves detailed pet profiles, enforcing the inclusion of critical medical conditions and allergies.
- Document Handling: Manages the digital records of veterinary vaccination certificates.
- Daily Reporting: Receives and securely stores daily health logs (feeding, mood, potty status) submitted by caretakers for pets currently staying at the hotel.

9.4 Billing & Financial Services

Automates the hotel's revenue tracking to eliminate manual calculation errors.

- Dynamic Cost Calculation: Generates real-time invoice previews by automatically combining locked room rates with the cost of any used inventory or add-on services.
- Invoice Finalization: Generates official, itemized billing records for completed stays.
- Payment Processing: Records finalized transactions and updates invoice statuses upon successful payment via cash, card, or digital transfer.

9.5 Inventory & Analytics Services

Provides facility managers with data-driven insights and resource tracking.

- Executive Dashboard Aggregation: Compiles key performance indicators—such as total revenue, current occupancy rates, and most popular add-on services—into a single, summarized data feed.
- Resource Tracking: Monitors the catalog of facility supplies and automatically deducts item quantities as they are used during a pet's stay.
- Automated Alerts: Continuously monitors inventory levels and generates warnings when essential supplies drop below their designated minimum thresholds.

9.6 Notification & Alert Services

Ensures seamless communication between the system, the staff, and the pet owners.

- Internal Alerts: Generates and organizes internal system notifications for staff, such as alerting receptionists to newly created bookings.
- Status Tracking: Manages the "read" or "unread" status of all notifications to ensure no critical updates are missed.

10. Conclusion

10.1 Summary of Project Success

The Purrfect Stay project successfully transitioned a conceptual pet hotel management framework into a fully operational, centralized digital platform. By completing the development across all six core modules—ranging from Booking Management and Pet Care to Billing and Inventory—the system effectively eliminates the inefficiencies of traditional, paper-based workflows. The application now provides internal staff with a reliable, real-time environment to manage the complete lifecycle of a pet's stay, ensuring high standards of care and operational fluidity.

10.2 Technical and Architectural Achievements

From a strategic standpoint, the project validated the team's decision to utilize a database-centric architecture. By strictly adhering to normalization principles, the system achieved robust data integrity without relying on heavy backend processing.

- Business Logic at the Foundation: Complex financial calculations and data consistency rules were successfully automated directly within the database architecture. This ensures that essential information is never left blank, billing is mathematically guaranteed to be accurate, and operational categories remain strictly standardized.

- Optimized Performance: The backend server was engineered to handle complex data retrieval through highly optimized queries. This allows the system to intelligently gather and combine information from multiple interconnected areas in a single step, drastically reducing server load and frontend response times.

- Secure Access Control: The successful implementation of modern encryption and stateless authentication ensures that facility security is strictly enforced. Receptionists, caretakers, and managers are securely directed only to the dashboards and data they are explicitly authorized to handle.

10.3 Resolution of Anticipated Challenges

The system successfully addressed the primary operational risks identified during the initial analysis phase

- Financial and Operational Transparency: The implementation of an immutable tracking system ensures that all administrative actions and booking cancellations are permanently recorded. Furthermore, locking prices at the exact moment of reservation guarantees that all financial agreements remain transparent and tamper-proof.

- Data Completeness: The risk of missing medical records was mitigated by establishing strict mandatory fields for pet health profiles and daily care reports. This ensures that caretakers always have reliable, complete data before interacting with the animals.

- Resource Conflict Prevention: The logical structuring of reservations and physical room assignments ensures that real-time availability checks effectively prevent any concurrent double-booking scenarios.

10.4 Future Work and Scalability (Phase 2 Roadmap)

While the current release successfully delivers a comprehensive internal tool for hotel staff, the highly modular architecture establishes a strong foundation for immediate future enhancements

1. Customer-Facing Portal: Expanding the frontend interface to allow pet owners to self-register, request bookings online, and view their pet's daily care reports and photos directly.

2. Cloud Storage Integration: Upgrading the current file attachment modules (for vaccination certificates and daily care photos) to integrate seamlessly with dedicated, high-capacity cloud storage solutions.

3. Advanced Inventory Automation: Expanding the system's automated logic to instantly deduct supply quantities the moment an item is used, completely removing the need for manual stock adjustments by the staff.

4. Automated Notification Gateway: Integrating third-party communication services to transition the notification system from internal dashboard alerts to real-time mobile and text updates for pet owners.